

## Homework 4: Interval TOPSIS Implementation

### Part 1: Reproduce the example from the article

We implemented the Interval TOPSIS method according to the algorithm defined in the article “Extension of TOPSIS for Decision-Making Problems with Interval Data”. We tested the implementation on the dataset from the article (Table 3) and compared the results.

#### Our Implementation Results

| Alternative | Efficiency Interval | Mid-point | Half-width | Rank |
|-------------|---------------------|-----------|------------|------|
| City 4      | [0.663, 0.811]      | 0.737     | 0.074      | 1    |
| City 6      | [0.601, 0.781]      | 0.691     | 0.090      | 2    |
| City 5      | [0.552, 0.628]      | 0.590     | 0.038      | 3    |
| City 2      | [0.452, 0.628]      | 0.540     | 0.088      | 4    |
| City 3      | [0.243, 0.320]      | 0.282     | 0.039      | 5    |
| City 1      | [0.211, 0.264]      | 0.237     | 0.027      | 6    |

**Comparison with the article:** Our results exactly match the values reported in the article (Table 6). For instance, City 4 gets the exact same mid-point of 0.737 and ranks 1st.

### Part 2: Convert crisp data into intervals

Using the smartphone selection dataset from the previous assignment, we replaced each crisp value  $x$  with an interval  $[0.95x, 1.05x]$ .

#### Crisp TOPSIS Results

| Alternative          | Efficiency Interval | Mid-point | Half-width | Rank |
|----------------------|---------------------|-----------|------------|------|
| OnePlus 12           | [0.727, 0.727]      | 0.727     | 0.000      | 1    |
| Asus Zenfone 10      | [0.698, 0.698]      | 0.698     | 0.000      | 2    |
| Nothing Phone (2)    | [0.689, 0.689]      | 0.689     | 0.000      | 3    |
| Motorola Edge 50 Pro | [0.678, 0.678]      | 0.678     | 0.000      | 4    |
| Xiaomi 14 Pro        | [0.616, 0.616]      | 0.616     | 0.000      | 5    |
| Honor Magic 6 Pro    | [0.546, 0.546]      | 0.546     | 0.000      | 6    |
| iPhone 15 Pro        | [0.498, 0.498]      | 0.498     | 0.000      | 7    |
| Google Pixel 8 Pro   | [0.408, 0.408]      | 0.408     | 0.000      | 8    |
| Samsung S24 Ultra    | [0.332, 0.332]      | 0.332     | 0.000      | 9    |
| Sony Xperia 1 V      | [0.233, 0.233]      | 0.233     | 0.000      | 10   |

#### Interval TOPSIS Results ( $\pm 5\%$ uniform variation)

| Alternative | Efficiency Interval | Mid-point | Half-width | Rank |
|-------------|---------------------|-----------|------------|------|
|-------------|---------------------|-----------|------------|------|

|                      |                |       |       |    |
|----------------------|----------------|-------|-------|----|
| OnePlus 12           | [0.568, 0.883] | 0.725 | 0.158 | 1  |
| Asus Zenfone 10      | [0.574, 0.795] | 0.684 | 0.110 | 2  |
| Nothing Phone (2)    | [0.602, 0.753] | 0.678 | 0.076 | 3  |
| Motorola Edge 50 Pro | [0.579, 0.755] | 0.667 | 0.088 | 4  |
| Xiaomi 14 Pro        | [0.478, 0.792] | 0.635 | 0.157 | 5  |
| Honor Magic 6 Pro    | [0.424, 0.731] | 0.578 | 0.154 | 6  |
| iPhone 15 Pro        | [0.391, 0.668] | 0.529 | 0.139 | 7  |
| Google Pixel 8 Pro   | [0.324, 0.565] | 0.445 | 0.120 | 8  |
| Samsung S24 Ultra    | [0.274, 0.447] | 0.360 | 0.086 | 9  |
| Sony Xperia 1 V      | [0.174, 0.404] | 0.289 | 0.115 | 10 |

**Comparison:** The global ranking order remains identical to the crisp version, which is expected since we applied a uniform proportional scaling to all parameters for all alternatives. However, the interval method provides a measure of uncertainty (the half-width). We can see that the OnePlus 12 has a significantly larger half-width (0.158) compared to the Nothing Phone (2) (0.076), indicating that the Nothing Phone’s efficiency score is much less sensitive to this 5% variation.

### Part 3: Construct expert-based intervals

We established the following expert-based intervals for each criterion: - **Price:**  $[MSRP - \$100, MSRP + \$50]$  to account for occasional discounts and market inflation/tax discrepancies. - **Battery:**  $[95\% \times Marketed, 100\% \times Marketed]$  to account for battery degradation over time and batch inconsistencies. - **Camera Score:**  $[Score - 3, Score + 3]$  to account for subjective variations and lighting conditions in testing. - **Performance Score:**  $[85\% \times Max, 100\% \times Max]$  to account for thermal throttling under heavy load.

#### Expert Interval TOPSIS Results

| Alternative          | Efficiency Interval | Mid-point | Half-width | Rank |
|----------------------|---------------------|-----------|------------|------|
| OnePlus 12           | [0.493, 0.956]      | 0.725     | 0.231      | 1    |
| Asus Zenfone 10      | [0.506, 0.883]      | 0.695     | 0.189      | 2    |
| Nothing Phone (2)    | [0.540, 0.828]      | 0.684     | 0.144      | 3    |
| Motorola Edge 50 Pro | [0.516, 0.833]      | 0.674     | 0.158      | 4    |
| Xiaomi 14 Pro        | [0.417, 0.847]      | 0.632     | 0.215      | 5    |
| Honor Magic 6 Pro    | [0.377, 0.754]      | 0.566     | 0.189      | 6    |
| iPhone 15 Pro        | [0.344, 0.712]      | 0.528     | 0.184      | 7    |
| Google Pixel 8 Pro   | [0.293, 0.579]      | 0.436     | 0.143      | 8    |
| Samsung S24 Ultra    | [0.249, 0.452]      | 0.351     | 0.102      | 9    |
| Sony Xperia 1 V      | [0.160, 0.408]      | 0.284     | 0.124      | 10   |

**Comparison:** When using expert intervals, the ranking remains consistent with the crisp method, but the efficiency intervals become significantly wider (larger half-widths) than in Part 2. This reflects the greater, more realistic uncertainty

modelled by the expert parameters, particularly the substantial performance drop (up to 15%) from thermal throttling.

For instance, OnePlus 12 remains the top choice based on the mid-point score, but its interval  $[0.493, 0.956]$  significantly overlaps with the Asus Zenfone 10  $[0.506, 0.883]$ . If a decision-maker is heavily pessimistic (meaning they look at the lower bound of efficiency), they might actually prefer the Nothing Phone (2), which guarantees a minimum efficiency of 0.540, higher than OnePlus 12's worst-case efficiency of 0.493.